ATLAS Second Mathematical Audit

DATA 3.0 / AMLD: TISO Repaired Edition

Review of repaired PDF, LaTeX, unchanged figures, and mathematical claims

October 16, 2026

SECOND-AUDIT VERDICT: MATHEMATICAL PASS WITH EXPLICIT MODEL CONDITIONS

No material mathematical defect found in the re-audited conditional optimization results. Calibration, the event process, and real AMLD performance remain unvalidated.

1. Integrity, scope, and what changed

Examined the original submitted regenerated ZIP and the repaired source/PDF. The repair is a *manuscript-only* action; it performed no AMLD run, no theorem-search experiment, no model fitting, and no executable protocol or parameter change. A programmatic section-heading comparison finds the original **19 sections in the same order**. SHA-256 comparison finds all **three figure assets byte-identical**. The repaired LaTeX compiles successfully twice without unresolved cross-references, compilation errors, or overfull-box warnings. It now renders as a 15-page PDF (the original was 13 pages), principally because new accounting and proof details were added.

Repairs are confined to existing sections:

- Sections 2, 4, 5, 8, 9: positive finite density denominator, precise KKT necessity/sufficiency and bound-active marginals, explicit online-regret feedback conditions, and finite-graph versus learned proof distance.
- Section 13: a globally conserved, event-identified credit convention and explicit warning that this convention *does not imply* stochastic independence or calibration.
- Section 14: separate Poisson and event-to-settlement hypotheses, limits on aggregate settle/not-settle prediction, and explicit nonclaims concerning class probabilities.
- Sections 15 and 18: rigorous strict-concavity criterion on the feasible set, and a new restricted static quantization guarantee, with accounting and dynamic-control boundaries.
- Abstract and conclusion: factual scope limited to the proven static surrogate, with untested claims clearly identified.

2. Audit matrix

Issue / theorem	Verdict	Re-audit conclusion
One-hot evidence classification (12)	**PASS**	Direct Boolean check: all eight triples yield exactly one reported state. U remains unresolved evidence, not a terminal certificate.
Event-credit conservation (13)	**PASS**	Nonnegative globally conserved shares imply total model credit $\sum\theta \le \sum_e \omega_e$ for the stated additive construction. Does not prove Poisson independence.
Poisson probability reading (14)	**CONDITIONAL**	$\Pr[N=0]=e^{-H}$ follows from $N\sim\text{Poisson}(H)$; the distribution and match to accepted settlement remain unverified assumptions.
Concavity, existence, uniqueness (15)	**PASS**	Correct negative-semidefinite Hessian. Revised sufficient uniqueness condition ensures every nontrivial feasible segment varies in a positive-weight coordinate.
KKT and water filling (4, 15)	**PASS**	Correct under stated static problem. Qualification needed for KKT necessity, not sufficiency. Single multiplier is not asserted with coupled constraints.
Numerical two-agent example (16)	**PASS**	$u_P = 0.79726745$, $u_D = 1.20273255$, $H = 0.95932985$, $F = 0.61685043$.
Contradiction complementarity (17)	**PASS**	At the origin Hessian eigenvalues are $\pm ab$, proving failure of joint concavity near the origin.
2ε proxy regret (7, 18)	**CONDITIONAL**	Correct comparison proof, but requires an actual uniform error guarantee on the same feasible set; none is established for AMLD.
Quantization bound (18)	**PASS**	New proof correct for nonnegative A, fixed marginal charges, feasible minima, and compatible quanta. Not valid by this proof for arbitrary coupled constraints or dynamic costs.
Dynamic real-world scheduler	NOT TESTED	No empirical, stochastic-process, integer-policy, or adaptive global optimality guarantee claimed or verified.

3. Independent checks of the principal mathematics

Set $H(u)=\sum_j W_j(1-e^{-k_j u_j})$, with $W_j\ge 0$, $k_j>0$, and $F(u)=1-e^{-H(u)}$. Its Hessian obeys

$$v^\top \nabla^2 F(u) v = -e^{-H(u)}\left[\sum_j W_j k_j^2 e^{-k_j u_j} v_j^2 + \left(v^\top \nabla H(u)\right)^2\right] \le 0.$$

This analytic identity proves concavity, independently of the probability interpretation. Existence follows by continuity/compactness; strict concavity holds on the feasible set when distinct feasible allocations cannot differ only along zero-weight coordinates. Stationarity of the equivalent objective H gives $W_j k_j e^{-k_j u_j} = \lambda c_j$ for positive-weight interior channels; the lower-bound clip and budget multiplier are correct.

New quantization result. On a static feasible region with entrywise nonnegative capacity matrix A and $c_j>0$, rounding u_j^* down to $m_j + q_j\lfloor (u_j^*-m_j)/q_j\rfloor$ preserves feasibility, because the resource inequalities are monotone. Since $0\le \partial_j F \le W_j k_j$, the objective loss is no more than $\sum_j W_j k_j q_j$. This is a genuine mathematical addition to the repaired manuscript, but is not a general rounding theorem for arbitrary A or an adaptive scheduler guarantee.

Finite reproducibility checks (not AMLD experiments). An accompanying deterministic script checked all 8 evidence triples; 5,000 random Hessians and finite-difference gradients (largest numerical Hessian eigenvalue 1.7×10^{-16}, consistent with roundoff); 3,000 event-share conservation examples; 4,000 monotone-feasibility rounding instances; 4,000 finite-set 2ε comparisons; and 1,000 contradiction-Hessian examples. All checks passed. The mathematical proofs, not this sampling, establish the universal statements.

4. What the repairs do not settle

R1. Calibration and predictive validity: OPEN. The map $\mathcal{K}_x$ has not been fitted and independently evaluated on predeclared held-out AMLD tasks. The optimized value is a valid *surrogate* regardless of empirical fit, not an established success rate.

R2. Joint event law and deduplication implementation: OPEN. Credit shares conserve attribution in the declared ledger. They do not prove independent opportunity arrival streams, handle every correlated reuse phenomenon, or implement canonical event equivalence in an actual system.

R3. Class-specific probabilities: NOT CLAIMED. Only aggregate terminal settlement probability is conditionally modeled. A sound joint model for P, R, I, K needs an additional event structure and consistency constraints. In particular known-consistent sound T precludes genuine K outcomes.

R4. Actual discrete/adaptive performance: OPEN. The static quantization theorem does not include policy-dependent future states, random verifier/communication overhead, early stopping, or coefficient refitting. Separate approved experimentation would be needed.

R5. Historical-source identity: NOT VERIFIED. The original six-page predecessor was not part of the supplied regenerated ZIP; preservation of the predecessor's *historical text* cannot be independently certified. The regenerated ZIP itself and its original 19-section order are the reproducible comparison baseline.

5. Final release recommendation

ATLAS recommendation: PASS for a carefully labeled conditional mathematical modeling paper. The prior audit's four high-priority clarification gaps have been repaired *at the manuscript level*: probability is expressly conditional, accounting is auditable, class-specific claims are withheld, and static-to-discrete scope is delimited with a theorem. No fatal inconsistency was found on re-audit. The paper should *not* be described as a validated AMLD optimizer, a statistically calibrated settlement forecaster, a proven dynamic controller, or experimental evidence of proof-discovery gains. No executable or experiment changes were authorized or made.